CBE 142 Crib sheet

Energy balance:

$$\frac{\partial E}{\partial t} = \dot{Q} + \dot{W}_s - F_{A0}\sum \Theta_i \tilde{c}_p (T-T_0) - F_{A0}X_A\left[\Delta H^\circ_{rxn} + \Delta\hat{c}_p(T-T_R)\right]$$

$$= \dot{Q} - \dot{W}_s + \underbrace{F_{A0}\sum\Theta_i(H_{i0}-H_i) - \Delta H_{Rx}F_{A0}X}_{\sum F_{i0}H_{i0} - \sum F_i H_i}$$

- $k = A_0 \exp\left(\frac{-E_a}{RT}\right) = k_1(T_2)\exp\left[\frac{E_a}{R}\left(\frac{1}{T_2}-\frac{1}{T}\right)\right]$
- $K_c = K_c(T_c)\exp\left[\frac{\Delta H^\circ_R}{R}\left(\frac{1}{T_2}-\frac{1}{T}\right)\right]$
- $\Delta H_{Rxn} = \Delta H^\circ_{Rxn} + \Delta\tilde{c}_p(T-T_R)$
- $S_D = \frac{r_D}{r_u} = \frac{F_D}{F_u}$; $\tilde{Y} = \frac{F_D}{F_{A0}-F_A}$; $Y = \frac{r_D}{-r_A}$

Adiabatic (Steady state; $W_s = 0$)

$$X^{EB} = \frac{\sum\Theta_i c_{p_i}(T-T_0)}{-\left[\Delta H^\circ_R(T_R)+\Delta c_p(T-T_R)\right]}$$

$$T = \frac{X\left[-\Delta H^\circ_R(T_R)\right] + \sum\Theta_i c_p T_0 + X\Delta c_p T_R}{\sum\Theta_i c_{p_i} + X\Delta c_p}$$

$\Delta c_p = 0$

↪ $\approx T_0 + \frac{X\left[-\Delta H^\circ_{Rx}(T_R)\right]}{\sum\Theta_i c_{p_i}}$

Batch

$$\frac{dT}{dt} = \frac{\overbrace{-\Delta H_{Rx}(-r_AV)}^{Q_g} - \overbrace{\dot{m}c_{p_c}(T-T_a)\left[1-\exp\left(\frac{-UA}{\dot{m}c_{p_c}}\right)\right]}^{Q_r}}{\sum N_i C_{p_i}}$$

$N_{A0}\frac{dX}{dt} = -r_AV$; Adiabatic → $Q_r = 0$

$\frac{dT}{dt} = \frac{Q_g - Q_r}{N_{A0}(\sum\Theta C_{p_i} + \Delta C_p X)}$; $\sum N_i c_{p_i} = N_{A0}\bar{c}_{p_0}$ ($\Delta_{cp} = 0$)

Semi-Batch

$$\frac{dT}{dt} = \frac{\dot{Q} - W_s - \sum F_{i0}(H-H_{i0}) + (-\Delta H_{rx})(-r_AV)}{\sum N_i c_{p_i}}$$

$\sum F_{i0}(H-H_{i0}) = \sum F_{i0}c_{p_i}(T-T_{i0})$ (No phase change)

When $\dot{W}_s = 0$, $Q_r = \dot{Q} - \sum F_{i0}(H-H_{i0})$

$UA(T-T_a) = \dot{m}c_{p_c}(T-T_a)\left[1-\exp\left(\frac{-UA}{\dot{m}c_{p_c}}\right)\right]$ (high flow rate of coolant)

$Q_g = V\sum_{i=1}^{q} r_{ij}\Delta H_{Rxij}$ (Multiple reactions)

CSTR

$$X^{EB} = \frac{\frac{UA}{F_{A0}}(T-T_a) + \sum\Theta_i c_{p_i}(T-T_0)}{\left[-\Delta H_R(T_R)\right]} = \frac{\left(\frac{UA}{F_{A0}} + \bar{c}_{p_0}\right)(T^*-T)}{\left[\Delta H^\circ_{rxn} + \Delta\tilde{C}_p(T-T_R)\right]}$$

$\alpha = \frac{UA}{\bar{c}_p F_{A0}}$; $T^* = \frac{T_0+\alpha T_a}{(1+\alpha)}$; $\bar{c}_{p_0} = \sum\Theta_i c_{p_i}$; $V = \frac{F_{A0}X}{-r_A(X,T)}$

$G(T) = -X\Delta H^\circ_{Rxn} = \frac{(r_AV)}{F_{A0}}\Delta H_{rxn}$; $R(T) = \bar{C}_{p_0}(1+\alpha)(T-T^*)$

$\Delta\tilde{c}_p = \Delta\hat{c}_p = \Delta c_p$

$$\hat{E}_{sy} = \sum N_iE_i = \sum N_iU_i = \sum N_i(H_i - PV_i) = \sum N_iH_i - P\sum \cancel{N_i}V_i$$

$$\therefore \dot{Q} - \dot{W}_s + \sum F_{i0}H_{i0}\big|_{in} - \sum F_iH_i\big|_{out} = \left[\sum N_i\frac{dH_i}{dt} + \sum H_i\frac{dN_i}{dt}\right]$$

$\frac{dN_i}{dt} = -\nu r_AV + F_{i0} - F_i$; $\frac{dH}{dt} = c_p\frac{dT}{dt}$; $\sum F_{i0}(H_i-H_{i0}) = \sum F_{i0}c_{p_i}(T-T_{i0})$

$$\Rightarrow \boxed{\frac{dT}{dt} = \frac{\dot{Q} - \dot{W}_s - \sum F_{i0}(H_i-H_{i0}) + (-\Delta H_{Rx})(-r_AV)}{\sum N_iC_{p_i}}}$$

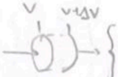
$N_i = C_iV$ (PFR)

For $\Delta_{cp}\approx 0$: $\sum N_ic_{p_i} = N_{A0}\sum\Theta_ic_{p_i}$; $\sum F_{i0}C_{p_i} = F_{A0}c_{p_i}$

PFR

$q - F_{A0}\frac{dT}{dV}(\bar{c}_{p_0} + \Delta\tilde{c}_p X_A) - F_{A0}\frac{dX_A}{dV}\left(\Delta H^\circ_{rxn} + \Delta\hat{c}_p(T-T_R)\right) = 0$; $F_{A0}\frac{dX}{dV} = -r_A(X,T)$

$q = Ua_v(T_a-T) = U\left(\frac{4}{D}\right)(T_a-T)$



$$\begin{cases} Q\Delta V + \sum F_iH_i\big|_V - \sum F_iH_i\big|_{V+\Delta V} = 0 \\ Ua(T_a-T) - \frac{d\sum(F_iH_i)}{dV} = 0\end{cases} \Rightarrow \frac{dT}{dV} = \frac{\overbrace{r_A\Delta H_{Rx}}^{Q_g} - \overbrace{Ua(T-T_a)}^{Q_r}}{\sum F_ic_{p_i}} \text{ (steady state)}$$

$\sum F_ic_{p_i} = F_{A0}(\sum\Theta_ic_{p_i} + \Delta C_pX)$

↓

Heat exchanges

$F\frac{dH}{dV} = Ua(T_a-T)$; $F = \frac{\dot{m}}{MW}$

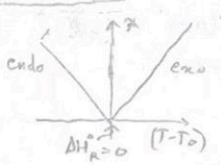
$\frac{dT}{dV} = \frac{Ua(T_a-T)}{Fc_p}$

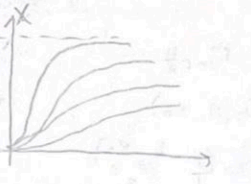
Co-current : $\frac{dT_a}{dV} = \frac{Ua(T-T_a)}{\dot{m}_c c_{p_c}}$

Counter-current : $\frac{dT_a}{dV} = \frac{Ua(T_a-T)}{\dot{m}_c c_{p_c}}$

PBR $\frac{dT}{dW} = \frac{r'_A\Delta H_R - \frac{Ua(T-T_a)}{\rho_b}}{\sum F_ic_{p_i}}$

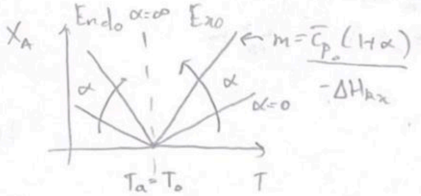
X vs. T graphs

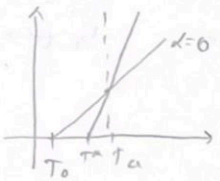




Adiabatic

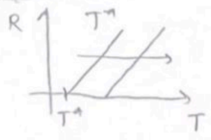
$\tau\uparrow$, $X_A^{MB}\uparrow$

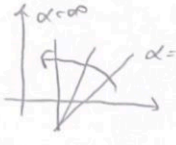


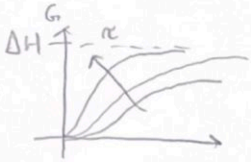


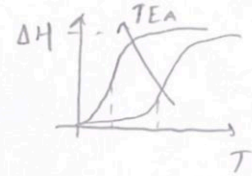
Non-Adiabatic

G vs. R graphs









Midterm 1

BSTR

$r_A V = \frac{dN_A}{dt}$

V≠cond.

$\frac{dN_A}{dt} = VdC_A + \frac{C_A dV}{dt}$

$N_{A0}\frac{dX}{dt} = (-r_A V)$

$\frac{t}{C_{A0}} = \int_0^{X_A}\frac{dX_A}{(-r_A)}$

PFR

$r_A = \frac{dF}{dV}$

$\nu = 0$ const.

$d\tau = \frac{dC_A}{r_A}$

$\frac{dX_A}{dV} = -\frac{r_A}{F_{A0}}$

$\frac{\tau}{C_{A0}} = \int_0^{X_A}\frac{dX_A}{(-r_A)}$

CSTR

$r_A = \frac{F_A - F_{A0}}{V}$

ν_0 const.

$V = \frac{\nu_0(C_{A0}-C_A)}{-r_A}$

$V = -\frac{F_{A0}X}{r_A}$

Mass Balance: $\frac{dm}{dt} = \dot m_{in} - \dot m_{out}$; $\dot m_i = (F_i)MW_i = ((C_i\nu))MW_i$

A F_{A0} $F_{A0}(\Theta - X_A)$
B F_{B0} $F_{A0}(\Theta_B - b/a\,X_A)$
C F_{C0} $F_{A0}(\Theta_C + c/a\,X_A)$
D F_{D0} $F_{A0}(\Theta_D + d/a\,X_A)$ $y_D^* F_T$

$\tau = \frac{V}{\nu_0}$; $F = C\cdot\nu$; $N = CV = F\cdot t$ In − Out + Gen. = Acc. (For 1 species)

Gases: $F_A = F_{A0}(1-X_A)$; $\nu = \nu_0\left(\frac{F_T}{F_{T0}}\right)\left(\frac{T}{T_0}\right)\left(\frac{P_0}{P}\right)$; $V = V_0\left(\frac{N_T}{N_{T0}}\right)\left(\frac{T}{T_0}\right)\left(\frac{P_0}{P}\right)$; $\frac{F_T}{F_{T0}} = 1+\varepsilon X_A$; $\varepsilon = y_A\delta$

$C_j = \frac{C_{j0}(\Theta_j + \nu_j X)}{(1+\varepsilon X)}\left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right) = C_{T0}\left(\frac{F_A}{F_T}\right)\left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right)$; $y_A = \frac{F_{A0}}{F_{T0}}$, $\delta = \left(\frac{d}{a}+\frac{c}{a}-\frac{b}{a}-\frac{a}{a}\right)$

Ergun: $\frac{dP}{dW} = -\alpha\frac{P_0^2}{2P}(1+\varepsilon X)\left(\frac{T}{T_0}\right) = -\frac{G}{\rho g_c D_p}\left(\frac{1-\gamma}{\gamma}\right)\left(\frac{150(1-\gamma)\mu}{D_p}+1.75G\right)$;

First order : $Da = \tau k$

$Da = \frac{-r_{A0}V}{F_{A0}} = \frac{k_1 C_{A0}V}{\nu_0 C_{A0}} = \frac{\text{Rxn rate}}{\text{Convect. rate}}$

- $Da < 0.1$ ↳ $X < 0.1$
- $Da > 10$ ↳ $X > 0.9$

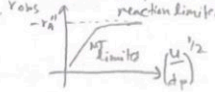
Membrane: $R_B = K_B' a_V(C_B - 0) \approx k_B C_B$; $\frac{dF_B}{dV} = r_B - R_B$

$Re = \frac{Ud_p}{(\mu/\rho)}$ $\frac{\text{inertial}}{\text{conv}}$; $Sc = \frac{(\mu/\rho)}{D_{AB}} = \frac{\text{momentum diff}}{\text{Mass diff.}}$; $Sh = \frac{k_c d_p}{D_{AB}} = \frac{\text{Conv. MT Resis.}}{\text{Diff MT Resis}}$

$k_c = \frac{D_{AB}}{\delta}$, $W_A = k_c(C_A^b - C_{As})$; $Sh = 2 + 0.6Re^{1/2}Sc^{1/3}$

$\eta = \frac{\text{Actual reaction rate}}{\text{Rxn rate if entire catalyst @}C_{As}, T_s} = \frac{r''_{obs}\text{ → actual}}{r''_{kinetics/C_{As}}}$

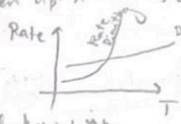
- For $Re\uparrow\uparrow$, $k_c \sim \left(\frac{U}{d_p}\right)^{1/2}$; $Re\downarrow\downarrow$, $k_c \sim \left(\frac{1}{d_p}\right)$



; $D_{eff} = D_{AB}\frac{\varepsilon}{\tau_p}$

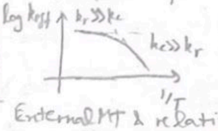
$D_{AB} > D_{eff}$

- When $d_p \gg \lambda$ → Bulk diffusivity
- When $d_p < \lambda$ → Knudsen diffusion
- When $d_p \sim \lambda$ → Configurational diff.



Diffusion ~ $T^{3/2}$

Reaction rate $= A\exp\left(\frac{E_a}{RT}\right)$



$\frac{1}{k_{eff}} = \frac{1}{k_c} + \frac{1}{k_r}$

External MT & relation to reaction rate

Porous catalyst

$(In@r) - Out|_{r+\Delta r} + gen_{\Delta r} = 0$

$W_A 4\pi r^2|_r - W_A\cdot 4\pi r^2|_{r+\Delta r} + (r_A'\rho_c 4\pi r_m^2\Delta r) = 0$

$\frac{d(W_A r^2)}{dr} - r_A'\rho_c r^2 = 0$; $r_A' = \underbrace{k_n'' S_a \rho_c}_{k_n} C_A^n$

$W_{A,r} = -D_e\frac{dC_A}{dr}$; $r = R \Rightarrow C_A = C_{As}$; $r = 0 \Rightarrow \frac{dC_A}{dr} = 0$

$\frac{d^2C_A}{dr^2} + \frac{2}{r}\frac{dC_A}{dr} - \frac{k_n}{D_e}C_A^n = 0$; $\psi = \frac{C_A}{C_{As}}$; $\lambda = \frac{r}{R}$

$\frac{d^2\psi}{d\lambda^2} + \frac{2}{\lambda}\frac{d\psi}{d\lambda} - \frac{k_n R^2 C_{As}^{n-1}}{D_e}\psi^n = 0$

$\Phi^2 = \frac{k_n R^2 C_{As}^{n-1}}{D_e} = \frac{\text{surface rxn rate}}{\text{Int. Diffusion rate}}$

$\Phi^2 \gg 1$:

- Strong internal MT limitation,

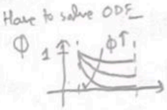
$\eta = \left(\frac{2}{n+1}\right)^{1/2}\frac{3}{\Phi_n} = \left(\frac{2}{n+1}\right)^{1/2}\frac{3}{R}\sqrt{\frac{D_e}{k_n}}C_{As}^{(1-n)/2}$; $\eta \propto \frac{1}{\Phi}$

$\Phi^2 \ll 1$:

- Reaction rate limited regime ($\eta \sim 1$)

$\Phi^2 \sim 1$:

- Have to solve ODE



$\varphi = \frac{C_A}{C_{As}} = \frac{1}{\lambda}\frac{\sinh(\Phi_1\lambda)}{\sinh(\Phi_1)}$

$\eta = \frac{3}{\Phi_1^2}(\Phi_1\coth\Phi_1 - 1)$ [Spherical, 1st order]

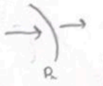
Imp SS balance :

1) $\eta k'' C_{As} = k_c(C_{Ab} - C_{As})a_c$

(Reaction rate per V) = (rate of external transport through BL)

2) $\underbrace{\Delta H_{rxn}(4\pi r^2)D_{eff}\frac{dC_A}{dr}}_{\text{Energy released in pellet}} = \underbrace{k_{eff}\frac{dT}{dr}(4\pi r^2)}_{\text{Energy conducted out of pellet}}$

3) $\underbrace{h(T_b - T_s)}_{\text{Ext. Heat transfer thru BL}} = \underbrace{k_c(C_{Ab} - C_{As})\Delta H_{rxn}}_{\text{Heat released by rxn in particle}}$



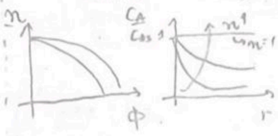
$-D_{eff}\Delta H_{rxn}\frac{dC_A}{dr}\Big|_{r=R} = -k_e\frac{dT}{dr}\Big|_{r=R}$

$\boxed{\frac{dT}{dC_A} = \frac{\Delta H_{rxn}D_{eff}}{k_{eff}}}$ Non-Isothermal EB

$\Phi \gg 1$

- $r''_{obs} = \eta r''_{kinetics} \Rightarrow C_{As}^{\frac{n+1}{2}} = \underbrace{C_{As}^{n'}}_{\text{measured}}$; $n' = \frac{n+1}{2}$ true ↳ measured/apparent
- $E_{app} = \frac{E_{kin}}{2}$; $k_{app} = \frac{k_2}{K_1}$
- $\frac{d(\ln k_{app})}{d(1/T)} = \frac{d(\ln k_2)}{d(1/T)} + \frac{d(\ln K_1)}{d(1/T)}$, $\frac{d(\ln k_{app})}{d(1/T)} = -\frac{E_{app}}{R}$

$\frac{d(\ln K_1)}{d(1/T)} = -\frac{\Delta H}{R}$



$r_{net} = \frac{k}{[L]}C_A^n$ (When two reactants on site)

External Film T gradients:

$\overset{\to}{T_s}))^{T_g}$ $h4\pi R^2(T_s - T_g) = -\frac{4}{3}\pi R^3(-r''_{A,obs})A_i\Delta H_R$ (EB)

$k_c(C_{Ag} - C_{As})4\pi R^2 = k_r C_{As}A_i\frac{4\pi R^3}{3}$ (MB)

∴ $Q_{rem} = Q_{gen}$

$(T_s - T_g) = \frac{RA_i(-\Delta H_R)}{3h}\frac{k_r k_c}{\left(\frac{A_i R}{3}k_r + k_c\right)}\cdot C_{Ag}$

$k_r \sim e^{-E/RT}$

$k_c \sim T^{3/2}$ (gas)

T_s gives us MSS

$-r_A\ [=]\ \frac{mol}{dm^3\cdot s}$ $(-r_A') = S_a(-r_A'')$

$(-r_A) = \rho_c(-r_A')$

$-r_A'\ [=]\ \frac{mol}{g_{cat}\cdot s}$ $(-r_A) = \rho_c S_A(-r_A'')$

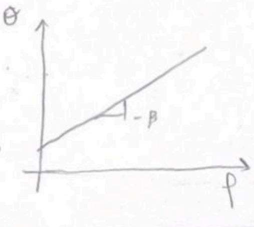
$-r_A''\ [=]\ \frac{mol}{m^2\cdot s}$

$\frac{\Delta T_{max}}{T_s} = \beta\left(1 - \frac{C_A(r=0)}{C_{As}}\right) \leq \beta$

$\theta = (1+\beta) - \beta\varphi_A$

$\theta = \frac{T}{T_s}$; $\varphi = \frac{C_A}{C_{As}}$

$\beta = \frac{-D_{eff}C_{As}\Delta H_R}{k_{eff}T_s}$



PBR

$A_c W_{Az}|_z - A_c W_{Az}|_{z+\Delta z} + r_A'\rho_b A_c\Delta z = 0$

$\Rightarrow -\frac{dW_{Az}}{dz} + r_A'\rho_b = 0$; $W_{Az} = -D_A\frac{dC_{Ab}}{dz} + UC_{Ab}$

$\Rightarrow \boxed{D_{AB}\frac{d^2C_{Ab}}{dz^2} - U\frac{dC_{Ab}}{dz} + r_A = 0}$

w/ catalysts

- $\frac{dF_A}{dW} = -r_A'$ [Design eq]

$W = \rho_b\cdot V$ $\rho_b = \rho_c(1-\varepsilon)$

WP & Mears:

- $C_{WP} = \Phi^2\eta = \frac{r'_{A,obs}\rho_c R^2}{D_e C_{As}}$

$C_{WP} \ll 1$ (No Internal diff limited)

$C_W \gg 1$ (Int. Diff limited)

- If $\frac{-r_A'\rho_b R_n}{k_c C_{Ab}} < 0.15$ (No ext. Diff. limit.)
- If $\left|\frac{-\Delta H_{Rx}(-r_A')\rho_b RE}{hT^2 R_g}\right| < 0.15$ (No T gradients)